



# Lesson Plan

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For

# Google Associate Cloud Engineer Training

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Course Code TGS-2023041024

TSC Cloud Computing · ICT-DIT-3020-1.1

Synchronous e-Learning

Conducted by Tertiary Infotech Academy Pte Ltd

UEN 201200696W

Version v8.0 · 12 September 2026

## Document Version Control Record

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                 | Author            |
|----------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------|
| 7.0            | Legacy release            | Original reference deck and course materials                                                                                | Tertiary Infotech |
| 8.0            | 12 September 2026         | Rebuilt Lesson Plan: current ACE domains, technical mechanisms, ten Activities, detailed verification and evidence guidance | Dr Alfred Ang     |

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## Course Delivery Plan

| Field      | Value                                    |
|------------|------------------------------------------|
| Course     | Google Associate Cloud Engineer Training |
| Code       | TGS-2023041024                           |
| TSC        | Cloud Computing · ICT-DIT-3020-1.1       |
| Delivery   | Synchronous e-Learning                   |
| Training   | 30 hours                                 |
| Assessment | WA 1 hour + PP 1 hour                    |
| Total      | 32 hours                                 |

## Learning Outcomes

- LO1: Set up and test Google Cloud solutions in accordance with security standards.
- LO2: Detect and resolve Google Cloud integration issues using technical solutions and database administration techniques.
- LO3: Conduct implementation tests against defined metrics and escalate unresolved Google Cloud issues with evidence.

## Instructional Strategy

The trainer alternates mechanism-led explanation, editable architecture and chart interpretation, live demonstration, guided Activity practice, evidence review and diagnostic questioning. Detailed procedures are delivered through the Learner Guide and Activity PDFs rather than copied onto slides.

## Day 1

| Time        | Block   | Content                                                     | Slides                            | Method                                            |
|-------------|---------|-------------------------------------------------------------|-----------------------------------|---------------------------------------------------|
| 09:00–09:20 | Admin   | Attendance, introductions, outcomes and assessment briefing | 2–21                              | Lecture · demonstration · practical · questioning |
| 09:20–11:20 | Topic 1 | Google Cloud and Cloud Computing                            | 22–58                             | Lecture · demonstration · practical · questioning |
| 11:20–13:00 | Topic 2 | GCP Basics and IAM                                          | 59–95                             | Lecture · demonstration · practical · questioning |
| 13:00–14:00 | Lunch   | Lunch break                                                 | Facilitated practice / assessment | Break                                             |
| 14:00–17:40 | Topic 3 | Virtual Networks and Activity 3                             | 96–132                            | Lecture · demonstration · practical · questioning |
| 17:40–18:00 | Recap   | Evidence review and cleanup                                 | Facilitated practice / assessment | Lecture · demonstration · practical · questioning |

Daily verification: trainer checks learner account/project scope, produced artifacts, negative-test evidence and cost cleanup before close.

## Day 2

| Time        | Block    | Content                           | Slides                            | Method                                            |
|-------------|----------|-----------------------------------|-----------------------------------|---------------------------------------------------|
| 09:00–12:15 | Topic 4  | Virtual Machines and Activity 4   | 135–171                           | Lecture · demonstration · practical · questioning |
| 12:15–13:00 | Practice | Recovery evidence review          | Facilitated practice / assessment | Lecture · demonstration · practical · questioning |
| 13:00–14:00 | Lunch    | Lunch break                       | Facilitated practice / assessment | Break                                             |
| 14:00–16:45 | Topic 5  | Storage, databases and Activity 5 | 172–208                           | Lecture · demonstration · practical · questioning |
| 16:45–18:00 | Topic 6  | Hybrid connectivity foundations   | 209–245                           | Lecture · demonstration · practical · questioning |

Daily verification: trainer checks learner account/project scope, produced artifacts, negative-test evidence and cost cleanup before close.

## Day 3

| Time        | Block   | Content                              | Slides                            | Method                                            |
|-------------|---------|--------------------------------------|-----------------------------------|---------------------------------------------------|
| 09:00–11:30 | Topic 7 | Managed services and Activity 7      | 248–284                           | Lecture · demonstration · practical · questioning |
| 11:30–13:00 | Topic 8 | Kubernetes foundations               | 285–321                           | Lecture · demonstration · practical · questioning |
| 13:00–14:00 | Lunch   | Lunch break                          | Facilitated practice / assessment | Break                                             |
| 14:00–16:30 | Topic 8 | GKE Activity 8 and scaling diagnosis | 285–321                           | Lecture · demonstration · practical · questioning |
| 16:30–18:00 | Topic 9 | Monitoring, logging and Activity 9   | 322–358                           | Lecture · demonstration · practical · questioning |

Daily verification: trainer checks learner account/project scope, produced artifacts, negative-test evidence and cost cleanup before close.

## Day 4

| Time        | Block    | Content                                     | Slides                            | Method                                            |
|-------------|----------|---------------------------------------------|-----------------------------------|---------------------------------------------------|
| 09:00–12:00 | Topic 10 | Load balancing, autoscaling and Activity 10 | 361–397                           | Lecture · demonstration · practical · questioning |
| 12:00–13:00 | Revision | Cross-domain evidence and practice exam     | Facilitated practice / assessment | Lecture · demonstration · practical · questioning |
| 13:00–14:00 | Lunch    | Lunch break                                 | Facilitated practice / assessment | Break                                             |
| 14:00–16:00 | Revision | Integrated troubleshooting and cleanup      | Facilitated practice / assessment | Lecture · demonstration · practical · questioning |
| 16:00–17:00 | WA       | Written Assessment K1–K5                    | Facilitated practice / assessment | Assessment                                        |
| 17:00–18:00 | PP       | Practical Performance A1–A6                 | Facilitated practice / assessment | Assessment                                        |

Daily verification: trainer checks learner account/project scope, produced artifacts, negative-test evidence and cost cleanup before close.

## Assessment Mapping

| Instrument            | Duration   | Criteria | Evidence                                                |
|-----------------------|------------|----------|---------------------------------------------------------|
| WA SAQ                | 60 minutes | K1–K5    | Five open-ended responses grounded in course mechanisms |
| Practical Performance | 60 minutes | A1–A6    | Three scenario tasks using Activity evidence patterns   |

## Activity Mapping

| Activity                                           | Topic | Criteria       | Duration | First slide |
|----------------------------------------------------|-------|----------------|----------|-------------|
| 01 Create a Safe Google Cloud Training Environment | 1     | A1, A2, K1, K3 | 75 min   | 55          |
| 02 Implement Least-Privilege IAM                   | 2     | A1, K1         | 75 min   | 92          |
| 03 Build and Diagnose a Custom VPC                 | 3     | A3, K4         | 90 min   | 129         |
| 04 Deploy and Recover a Compute Engine Web VM      | 4     | A3, K4         | 90 min   | 168         |
| 05 Operate Cloud Storage and Diagnose Cloud SQL    | 5     | A4, K5         | 105 min  | 205         |
| 06 Model Hybrid Connectivity and BGP Failure       | 6     | A4, K5         | 90 min   | 242         |
| 07 Build a Pub Sub to Cloud Run Event Pipeline     | 7     | A3, A4, K4     | 105 min  | 281         |

| Activity                                           | Topic | Criteria   | Duration | First slide |
|----------------------------------------------------|-------|------------|----------|-------------|
| 08 Deploy and Test a GKE Autopilot Workload        | 8     | A3, A4, A5 | 120 min  | 318         |
| 09 Create Monitoring Evidence and an Incident Pack | 9     | A5, K2     | 90 min   | 355         |
| 10 Load Test a Managed Instance Group and Escalate | 10    | A5, A6, K2 | 120 min  | 394         |

## Resources and Contingencies

Primary delivery uses Zoom with Microsoft Teams as backup. Learners require a laptop, supported browser, Google Account, Google Cloud console access and Cloud Shell or Google Cloud CLI. Trainers must provide a mock/sandbox path when corporate policy or billing prevents live provisioning. No learner is required to expose payment information or credentials.